

Closing Auction Trading

Predicting the AAPL closing print from L2 order-book dynamics

Group 5 – Cohort 1

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The Economic Thesis

The structural friction

25–30% of US daily equity volume clears in the 4:00 PM NASDAQ closing auction. The net imbalance is published only at 3:50 PM — but it leaves a directional trace in the L2 book before then.

Why the edge persists

- MOC orders become irrevocable at 3:58 PM — no free arbitrage.
- Market makers hold back quotes (adverse selection).
- Reading a 12-minute L2 window is a high barrier to entry.

The trade idea, in one line

Observe 3:15–3:45 PM, predict the next 12 minutes with Ridge, enter at 3:45 PM, exit by 3:57 PM — before the auction prints.

The Closing Auction Mechanism

How the auction works

At **4:00 PM**, NASDAQ sets a single clearing price that fills all Market-on-Close (MOC) orders at once.

Institutions submit MOC orders all day — but the official imbalance feed is not published until **3:50 PM**.

Before 3:50 PM, that pressure has nowhere to go except into the continuous L2 book.

Directional logic

- Net **buy** imbalance $\Rightarrow$ price drifts **up** into 4:00 PM
- Net **sell** imbalance $\Rightarrow$ price drifts **down**

What this looks like in the L2 book

<i>Signal we observe</i>		<i>What it tells us</i>
Bid-side	depth	Institutions positioning ahead of buy-MOC
grows early		
Mid-price	trends	Accumulated flow absorbed in the session
steadily		
Move speeds up	near 3:45 PM	Urgency rising as the MOC cutoff nears
Ask quotes disappearing		Makers pulling exposure before clearing

Dataset

Symbol	AAPL
Source	Databento — XNAS.ITCH
Schema	MBP-10 (10 bid/ask levels)
Period	Sep 27, 2024 – May 15, 2025
Days	158 trading days
Split	90 train / 68 OOS

AAPL microstructure

- Spread $\approx 1\text{¢}$ on $\approx \$220 \Rightarrow$ round-trip cost ≈ 0.9 bps.
- Microprice: $P^\mu = \frac{P_a q_b + P_b q_a}{q_b + q_a}$.

Backtest discipline

Temporal split, no leakage. Weights frozen before OOS. Threshold set on economics, not tuned.

Feature Engineering: 5 L2 Features (3:15–3:45 PM)

Feature	Window	Economic intuition
<code>imb_early</code>	3:15–3:35	Exp-decay depth imbalance — sustained pre-positioning
<code>imb_late</code>	3:35–3:45	Same metric, final 10 min — where late MOC flow concentrates
<code>mom_30</code>	Full 30 min	Microprice return (bps), clipped — directional anchor
<code>mom_accel</code>	Last 15 min	$\text{mom}_{15} - \text{mom}_{30}$ — late acceleration, near-orthogonal to <code>mom_30</code>
<code>cancel_imb</code>	Full window	Ask-vs-bid cancellations — liquidity pulled pre-auction

- All features standardised (zero-mean, unit-variance on *training* days).
- `cancel_imb` replaces the earlier OFI feature: `action="C"` records exist every day, while trade records had inconsistent coverage.

The Model: Ridge Regression

Signal

$$\begin{aligned}\hat{y}_t = & \beta_0 + \beta_{ie} z(\text{imb_early}) + \beta_{il} z(\text{imb_late}) \\ & + \beta_{30} z(\text{mom_30}) + \beta_{ac} z(\text{mom_accel}) \\ & + \beta_{ci} z(\text{cancel_imb})\end{aligned}$$

Target: 12-minute mid-price return, in bps.

Why Ridge at $\alpha = 15$

- OLS unstable: $n = 90$, correlated features.
- L2 penalty trades a little bias for much less variance.
- $\alpha = 15$ keeps $\max |\hat{\beta}| \leq 3.5$.

Model quality

In-sample $R^2 = 0.096$; hit rate 55.6%.

Fitted Coefficients & Interpretation

Coefficients (standardised)

Feature	Coeff	Sign
Intercept	+0.46	—
imb_early	+0.46	↑
imb_late	−0.78	↓
mom_30	+3.52	↑↑
mom_accel	+2.99	↑↑
cancel_imb	−0.21	↓

Reading the signs

- **Momentum dominates:** a price trending and accelerating into the close tends to continue.
- **imb_late negative:** heavy late bid depth signals dealers absorbing sell-MOC — bearish.
- **cancel_imb negative:** ask-side cancellations precede downward auction pressure.

Entry (3:45 PM) — trade if both

- 1 $|\hat{y}_t| \geq 1.5$ bps (clears the cost buffer).
- 2 Spread $\leq 3\phi$ (liquidity filter).

BUY if $\hat{y}_t > 0$ (fill at ask); SELL if $\hat{y}_t < 0$ (fill at bid).

Position sizing

$$\text{qty} = \text{round}(|\hat{y}_t|/3) \in [1, 5]$$

Adaptive exit (checked every 15 s)

- Adverse ≥ 12 bps $\Rightarrow$ stop-loss.
- Favourable ≥ 22 bps $\Rightarrow$ take-profit.
- Otherwise $\Rightarrow$ scheduled exit 3:57 PM.

Asymmetric payoff

$$\frac{\text{TP}}{\text{SL}} = \frac{22}{12} = 1.83, \quad \text{break-even WR} = 35.3\%$$

Full spread crossed per round-trip — no midpoint fills.

In-Sample Results (Sep 2024 – Feb 2025)

Metric	Value
Net PnL (1-share)	\$6.45
Ann. return	\$18.06 / yr
Ann. volatility	\$4.89 / yr
Sharpe ratio	3.70
Win rate	54.5% (36/66)
Break-even WR	35.3%
Max drawdown	−\$1.23
Days traded	66 / 90

Win rate beats break-even

Expected PnL per trade:

$$0.545 \times 22 - 0.455 \times 12 = +6.5 \text{ bps}$$

Caveat on the IS Sharpe

Amplified by variable sizing — a 5-share take-profit day counts 5× a 1-share day. True quality: $R^2 = 0.096$, hit rate 55.6%.

Out-of-Sample Results (Feb – May 2025)

Metric	Train	OOS
Net PnL (\$)	6.45	11.39
Ann. return (\$/yr)	18.06	42.21
Ann. vol (\$/yr)	4.89	8.35
Sharpe	3.70	5.05
Win rate	54.5%	59.5%
Max drawdown	−1.23	−3.04
Days traded	66/90	42/68

Why OOS Sharpe exceeds IS

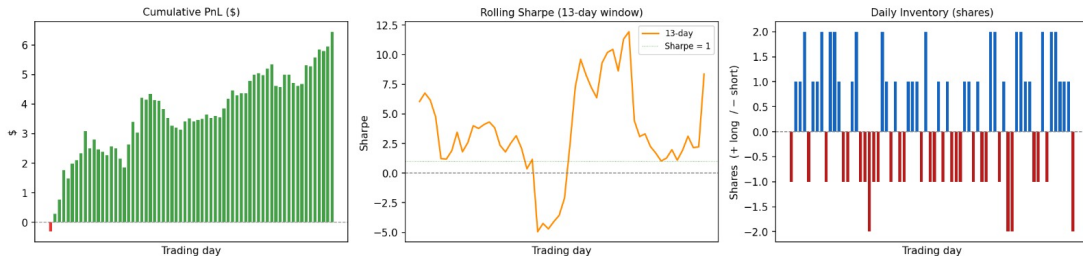
The Mar–Apr 2025 tariff sell-off (AAPL ~ \$250 → \$170) meant larger moves when the signal fired — raising per-trade PnL.

Honest caveat

42 OOS days; $t \approx 2.3$ ($p < 0.05$) but the CI is wide. Stop-loss share rose to 29% (vs 6% IS).

Strategy Diagnostics (In-Sample)

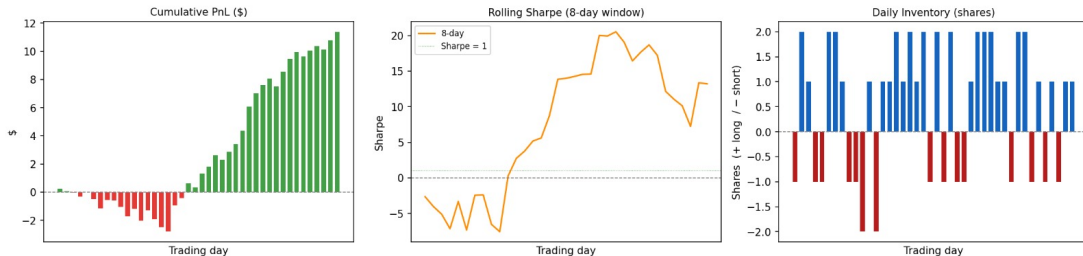
Closing Auction Strategy — In-Sample — Train Set



Cumulative PnL trends up steadily; rolling 13-day Sharpe sits mostly above 1; daily inventory stays within ± 2 shares — flat by 3:57 PM every day.

Strategy Diagnostics (Out-of-Sample)

Closing Auction Strategy — OOS — Test Set



Same pattern out-of-sample: PnL dips early, then turns positive and trends up; rolling 8-day Sharpe climbs well above 1; inventory stays within ± 2 shares.

Beta & Alpha: Is the Sharpe Real?

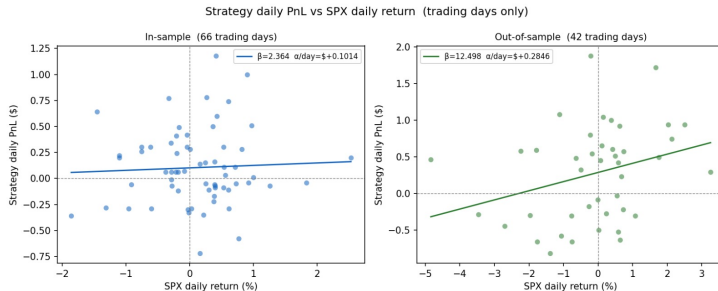
Designed to be market-neutral

- **12-min hold:** the market barely moves in 12 minutes — beta needs sustained exposure.
- **Zero overnight:** flat by 3:57 PM — no gap risk.
- **But a long tilt remains:** 27 long / 15 short in OOS (64/36) leaves residual market exposure.

What the data shows

IS beta ≈ 0 as designed. OOS beta is positive and significant — the long tilt got priced in during the high-dispersion tariff regime, when market direction and auction-flow pointed the same way.

Beta & Alpha: PnL Regressed on SPX Returns



	IS	OOS
β (\$/SPX ret.)	+2.58	+12.50
$t(\beta)$	0.46	2.13
p	0.64	0.03
α (\$/yr)	+26.0	+71.7
R^2	0.003	0.089

- **IS:** $\beta \approx 0$ — market-neutral, as designed.
- **OOS:** $\beta > 0$, significant — 1% SPX $\approx$ +\$0.13/day, from the long tilt in the tariff regime.
- $R^2 = 8.9\%$: $\sim 91\%$ of PnL is idiosyncratic alpha. A risk to disclose, not a design flaw.

Risk: Regimes That Hurt the Strategy

- **Macro event 3:45–3:57:** FOMC / earnings / block print after entry overwhelms the signal — stop-loss fires.
- **Late MOC flip:** a large opposing order reverses the imbalance; the signal is stale.
- **Thin liquidity:** a single order moves `imb_late` and makes features noisy.

Decay monitor

Flag when trailing 30-day $R^2 < 5\%$ or the hit rate nears the 35.3% break-even.
Retrain quarterly.

Risk: Capacity

Deployment	Shares / trade	Est. market impact
Current (1–5 sh)	≤ 5	Nil
\$10M	~45K	< 0.5 bps
\$100M	~450K	2–4 bps (erodes edge)
\$1B	~4.5M	> 20 bps (breaks)

Capacity break $\approx$ \$50–100M

Where market impact at entry reaches the 1.5 bps threshold. Scaling further requires spreading entry across time, limit orders, or a basket of liquid names.

What we got right

- Weights frozen before any test data.
- Orders fill at actual bid/ask.
- Threshold set on economics, not OOS.
- AAPL chosen *a priori* for liquidity.

Honest limitations

- IS Sharpe overstated by variable sizing.
- Feature iteration done in-sample.
- Small OOS sample (42 days); CI wide.
- Single ticker — generalisation untested.

Summary results

	IS	OOS
Sharpe	3.70	5.05
Win rate	54.5%	59.5%
Net PnL	\$6.45	\$11.39
R^2	0.096	—

Thesis — supported

- Falsifiable edge: $R^2 = 0.096$, hit rate 55.6%.
- Structural friction $\Rightarrow$ slow decay.
- OOS Sharpe exceeds IS in a volatile regime.

Limitations

- Capacity $\approx$ \$50–100M.
- Single ticker; 42 OOS days.

Thank you. Questions welcome.